

Jianhao Zhang, Ph.D.

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Research Interests

Clouds are fundamental to Earth's climate system, shaping both the hydrological cycle through precipitation and the global energy balance through their reflective (cooling) and heat-trapping (warming) properties. The processes governing cloud behavior span an enormous range of scales—from microscopic droplet formation to the evolution of weather systems hundreds of kilometers across—making clouds one of the least constrained components in predicting weather and climate.

My research characterizes and quantifies how clouds respond to changes in aerosol loading and meteorological conditions driven by human activity and natural climate variability. Building on this, I seek to understand how these cloud responses shape the increasing Earth's energy imbalance and constrain the viability and risks of proposed climate interventions. To address these multiscale questions, I leverage satellite and field observations, atmospheric and Earth system modeling, and data-driven approaches to extract physical understanding of the processes linking clouds to a changing climate.

Professional Appointment

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|------------------------|---|
| May 2026 – present | 📌 Research Scientist III , NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder |
| May 2023 – Apr 2026 | 📌 Research Scientist II , NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder |
| Sept. 2021 – Apr. 2023 | 📌 Research Scientist I , NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder |
| Sept. 2020 – Aug. 2021 | 📌 NRC Postdoc fellowship hosted by Graham Feingold at NOAA Chemical Sciences Laboratory |

Education

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|------|---|
| 2020 | 📌 Ph.D., University of Miami, Miami, FL, USA in Meteorology & Physical Oceanography.
Dissertation title: <i>The interactions between light-absorbing smoke and marine boundary layer clouds over the remote southeast Atlantic.</i>
Committee: <i>Paquita Zuidema (Chair), Brian Mapes, Brian Soden, Cassandra Gaston, David Turner & Takanobu Yamaguchi</i> |
| 2014 | 📌 B.S., Florida State University, Tallahassee, FL, USA in Meteorology.
<i>Magna cum laude; Minor in Mathematics.</i> |

Awards, Fellowships, & Honors

- 📌 **NOAA Boulder Outreach Gold Star Award, 2025**
- 📌 **Reflective Fellowship, 2025**, Reflective Inc.
- 📌 **National Research Council Research Associateship Fellowship, 2020**, National Academies of Sciences, Engineering, and Medicine
- 📌 **NASA's Group Achievement Awards** (ACTIVATE Team, 2023; ORACLES Team, 2019)
- 📌 **Finalist of University of Miami Best Ph.D. Dissertation, 2020**
- 📌 **University of Miami Graduate Fellowship, 2014** (highest award of the graduate school)

Funded Grants

2026-ongoing	■ Investigator: Assessing the detectability of small-scale stratospheric aerosol injections (DoW ESTCP Natural Hazards Project NH26-9821)
2025-2026	■ P. I.: A Framework for Assessing SRM Detectability for Informed Decision-Making (Reflective , \$162,768)
2023-2026	■ Co-I.: Aerosol-Cloud Interactions Centered on MAGIC: Insights from Measurements and Lagrangian Large Eddy Simulation (DOE Atmospheric System Research, \$670,445)
2023-2024	■ Co-P.I.: Exploring Aerosol-Cloud Interactions in Geophysical Variable Spaces using NASA-ACTIVATE Observations (NASA Earth Venture Suborbital-3, \$350,000)
2020-2023	■ Co-I.: Evaluating Biases in Aerosol-Cloud Interaction Metrics using ARM Data and Models (DOE Atmospheric System Research, \$510,472)
2020-2021	■ P. I.: National Research Council Fellowship Award (National Academies of Sciences, Engineering, Medicine , \$62,000)
2020-ongoing	■ NOAA Earth's Radiation Budget Program (internally funded)

Publications

Peer-reviewed

- 21 M. S. Diamond, A. S. Ackerman, A. Baró Pérez, et al., “Drizzle drives model differences in the Southeastern Atlantic Stratocumulus Transitions with Aerosol-Rain-Radiation interactions (SEA STARR) large eddy simulation intercomparison project,” *Atmos. Chem. Phys.*, 2026, [under review](#).
- 20 T. Dror, **J. Zhang**, and G. Feingold, “Amazon darkening mitigated by all-sky biophysical feedbacks,” *Nature Communications*, 2026, [under review](#).
- 19 **J. Zhang**, J. Gristey, and G. Feingold, “Earth’s East-West albedo symmetry,” *Nature*, vol. 654, pp. 676–682, 2026. [DOI: 10.1038/s41586-026-10624-2](#).
- 18 **J. Zhang**, D. Painemal, T. Dror, J.-S. Lim, A. Sorooshian, and G. Feingold, “Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite,” *Atmos. Chem. Phys.*, vol. 26, no. 9, pp. 6015–6034, 2026. [DOI: 10.5194/acp-26-6015-2026](#).
- 17 G. Feingold, F. Glassmeier, **J. Zhang**, and F. Hoffmann, “Opinion: Inferring process from snapshots of cloud systems,” *Atmos. Chem. Phys.*, vol. 25, no. 18, pp. 10 869–10 885, 2025. [DOI: 10.5194/acp-25-10869-2025](#).
- 16 **J. Zhang**, Y.-S. Chen, E. Gryspeerdt, T. Yamaguchi, and G. Feingold, “Radiative forcing from the 2020 shipping fuel regulation is large but hard to detect,” *Commun. Earth Environ.*, vol. 6, no. 18, pp. 1–11, 2025. [DOI: 10.1038/s43247-024-01911-9](#).
- 15 Y.-S. Chen, **J. Zhang**, F. Hoffmann, T. Yamaguchi, F. Glassmeier, X. Zhou, and G. Feingold, “Diurnal evolution of non-precipitating marine stratocumuli in a large-eddy simulation ensemble,” *Atmos. Chem. Phys.*, vol. 24, no. 22, pp. 12 661–12 685, 2024. [DOI: 10.5194/acp-24-12661-2024](#).
- 14 G. Feingold, V. Ghate, L. M. Russell, et al., “Physical science research needed to evaluate the viability and risks of marine cloud brightening,” *Science Advances*, vol. 10, no. 12, eadi8594, 2024. [DOI: 10.1126/sciadv.adi8594](#).
- 13 **J. Zhang**, Y.-S. Chen, T. Yamaguchi, and G. Feingold, “Cloud water adjustments to aerosol perturbations are buffered by solar heating in non-precipitating marine stratocumuli,” *Atmos. Chem. Phys.*, vol. 24, no. 18, pp. 10 425–10 440, 2024. [DOI: 10.5194/acp-24-10425-2024](#).
- 12 C. Howes, P. E. Saide, H. Coe, et al., “Biomass-burning smoke’s properties and its interactions with marine stratocumulus clouds in WRF-CAM5 and southeastern Atlantic field campaigns,” *Atmos. Chem. Phys.*, vol. 23, no. 21, pp. 13 911–13 940, 2023. [DOI: 10.5194/acp-23-13911-2023](#).

- 11 **J. Zhang** and G. Feingold, “Distinct regional meteorological influences on low-cloud albedo susceptibility over global marine stratocumulus regions,” *Atmos. Chem. Phys.*, vol. 23, no. 2, pp. 1073–1090, 2023. [DOI: 10.5194/acp-23-1073-2023](https://doi.org/10.5194/acp-23-1073-2023).
- 10 P. A. Barrett, S. J. Abel, H. Coe, et al., “Intercomparison of airborne and surface-based measurements during the clarify, oracles and lasic field experiments,” *Atmos. Meas. Tech.*, vol. 15, no. 21, pp. 6329–6371, 2022. [DOI: 10.5194/amt-15-6329-2022](https://doi.org/10.5194/amt-15-6329-2022).
- 9 M. S. Diamond, P. E. Saide, P. Zuidema, et al., “Cloud adjustments from large-scale smoke–circulation interactions strongly modulate the southeastern atlantic stratocumulus-to-cumulus transition,” *Atmos. Chem. Phys.*, vol. 22, no. 18, pp. 12 113–12 151, 2022, ([ACP highlight](#)). [DOI: 10.5194/acp-22-12113-2022](https://doi.org/10.5194/acp-22-12113-2022).
- 8 **J. Zhang**, X. Zhou, T. Goren, and G. Feingold, “Albedo susceptibility of northeastern pacific stratocumulus: The role of covarying meteorological conditions,” *Atmos. Chem. Phys.*, vol. 22, no. 2, pp. 861–880, 2022. [DOI: 10.5194/acp-22-861-2022](https://doi.org/10.5194/acp-22-861-2022).
- 7 **J. Zhang** and P. Zuidema, “Sunlight-absorbing aerosol amplifies the seasonal cycle in low-cloud fraction over the southeast atlantic,” *Atmos. Chem. Phys.*, vol. 21, no. 14, pp. 11 179–11 199, 2021. [DOI: 10.5194/acp-21-11179-2021](https://doi.org/10.5194/acp-21-11179-2021).
- 6 X. Zhou, **J. Zhang**, and G. Feingold, “On the importance of sea surface temperature for aerosol-induced brightening of marine clouds and implications for cloud feedback in a future warmer climate,” *Geophys. Res. Lett.*, vol. 48, no. 24, e2021GL095896, 2021. [DOI: https://doi.org/10.1029/2021GL095896](https://doi.org/10.1029/2021GL095896).
- 5 S. J. Abel, P. A. Barrett, P. Zuidema, **J. Zhang**, M. Christensen, F. Peers, J. W. Taylor, I. Crawford, K. N. Bower, and M. Flynn, “Open cells exhibit weaker entrainment of free-tropospheric biomass burning aerosol into the south-east Atlantic boundary layer,” *Atmos. Chem. Phys.*, vol. 20, no. 7, pp. 4059–4084, 2020. [DOI: 10.5194/acp-20-4059-2020](https://doi.org/10.5194/acp-20-4059-2020).
- 4 **J. Zhang** and P. Zuidema, “The diurnal cycle of the smoky marine boundary layer observed during August in the remote southeast Atlantic,” *Atmos. Chem. Phys.*, vol. 19, no. 23, pp. 14 493–14 516, 2019, ([ACP highlight](#)). [DOI: 10.5194/acp-19-14493-2019](https://doi.org/10.5194/acp-19-14493-2019).
- 3 A. S. Chandra, P. Zuidema, S. Krueger, A. Kochanski, S. P. de Szoek, and **J. Zhang**, “Moisture distributions in tropical cold pools from equatorial Indian ocean observations and cloud-resolving simulations,” *J. Geophys. Res. Atmos.*, vol. 123, no. 20, pp. 11, 445–11, 465, 2018. [DOI: 10.1029/2018JD028634](https://doi.org/10.1029/2018JD028634).
- 2 **J. Zhang**, P. Zuidema, D. D. Turner, and M. P. Cadeddu, “Surface-based microwave humidity retrievals over the equatorial Indian ocean: Applications and challenges,” *J. Appl. Meteor. Climatol.*, vol. 57, no. 8, pp. 1765–1782, 2018. [DOI: 10.1175/JAMC-D-17-0301.1](https://doi.org/10.1175/JAMC-D-17-0301.1).
- 1 P. Zuidema, A. J. Sedlacek III, C. Flynn, S. Springston, R. Delgadillo, **J. Zhang**, A. C. Aiken, A. Koontz, and P. Muradyan, “The Ascension island boundary layer in the remote southeast Atlantic is often smoky,” *Geophys. Res. Lett.*, vol. 45, no. 9, pp. 4456–4465, 2018. [DOI: 10.1002/2017GL076926](https://doi.org/10.1002/2017GL076926).

Other Publications

- 3 **J. Zhang** and G. Feingold, “Physical Science of Marine Cloud Brightening: Knowledge and Gaps,” in *Topical Group on the Physics of Climate*, ([invited article](#)), American Physical Society, October 2024, pp. 1–4. [URL: https://engage.aps.org/gpc/resources/newsletters](https://engage.aps.org/gpc/resources/newsletters).
- 2 D. Painemal, **J. Zhang**, and R. Ferrare, “Observational needs for reducing uncertainties in the quantification of aerosol-cloud interactions and radiative forcing,” white paper, A response to the Request for Information of the Decadal Survey for Earth Science and Applications from Space 2028–2037 (ESAS 2028), 2026.
- 1 G. Feingold, V. Ghatge, L. M. Russell, et al., “DOE-NOAA Marine Cloud Brightening Workshop,” in *U.S. Department of Energy and U.S. Department of Commerce NOAA, DOE/SC-0207; NOAA Technical Report OAR ESRL/CSL-1*, 2022, pp. 1–33.

Teaching, Mentoring, & Outreach

Teaching	■ Aerosol-Cloud-Climate Interactions as <i>Guest lecturer</i>, TU Delft (June 2024) ■ Introduction to Weather and Climate as <i>Teaching Assistant</i>, University of Miami (2015) ■ AP calculus and undergrad statistics as <i>Math Tutor</i>, Miami (2015-2020)
Mentoring	■ Tyler Tatro, Ph.D. student at University of Miami (2022-present) ■ Alexander J. Thompson, Research Scientist at NOAA/PSL (2024-2025) ■ Danyan Leng, Ph.D. student at University of Colorado Boulder (2025-2026) ■ Daisy Kerr, NOAA Hollings Scholar, Physics undergrad at UC Berkeley (Summer 2026)
Outreach	■ Guest Speaker, The Program for Exploratory Studies at CU Boulder (Apr 2026) ■ CIRES Speaker, CIRES Science Pathways Program (2025-) ■ Scientist Panel, Environmental studies class at Boulder High School (2025) ■ Panelist, CIRES Graduate Student Workshop (2025) ■ Invited Scientist, Boulder Valley Rotary Club Weekly Discussion (2025) ■ Evaluator, Research Experience for Community College Students Symposium 2024; GLOBE International Virtual Science Symposium 2022; Climate Literacy and Energy Awareness Network 2022, AGU Outstanding Student Presentation Award 2020 ■ Science vetting, Children's book <i>UP, UP HIGH</i>

Invited Presentations

May 2026	■ Libera Science Team Meeting, Berkeley, CA, USA , <i>Earth's East-West Albedo Symmetry</i> . (presented virtually)
Mar. 2026	■ NOAA Chemical Sciences Laboratory Seminar, Boulder, CO, USA , <i>A cloudy story of Earth's radiation budget</i> .
May 2025	■ AGU AS Early-Career Seminar, Online , <i>Global climate and air quality implications of regional emission shift</i> .
Apr. 2025	■ Boulder Valley Rotary Club Weekly Meeting, Boulder, CO, USA , <i>How to measure the shade of clouds that are no longer there</i> .
	■ Climate Dynamics & Impacts, Vecchi/Soden Joint Group Meeting, Princeton University and University of Miami, Online , <i>Large radiative forcing from the 2020 shipping fuel regulation is hard to detect: Implications for Marine Cloud Brightening</i> .
	■ University of Washington Inaugural MCB Program Workshop, Leavenworth, WA, USA , Session co-lead and presenter , <i>Identifying conditions amenable to cloud brightening/MCB</i> .
	■ NOAA Science Seminar Series, Online , <i>AI Applications in Earth System and Climate Science: Aerosols and Air Quality</i> .
Nov. 2024	■ NOAA Advancing Innovative Research Seminar Series, Online , <i>Large radiative forcing from the 2020 shipping fuel regulation is hard to detect</i> .
Jun. 2024	■ TU Delft, Delft, Netherlands , <i>On the viability of Marine Cloud Brightening: Albedo susceptibility, cloud adjustment, and detectability</i> .
Oct. 2023	■ Brookhaven National Laboratory, Long Island, NY, USA , <i>Aerosol-cloud interactions in marine warm clouds and implications for Marine Cloud Brightening</i> .
Dec. 2022	■ 2022 AGU Fall Meeting, Chicago, IL, USA , <i>On the Conditionality of Marine Low Cloud Albedo Susceptibility: from Meteorological Conditions to Spatiotemporal Scales</i> .

Leadership & Service

Grant Reviewer	■ Department of Energy (DOE) Atmospheric System Research (ASR) University of Colorado AB Nexus Program NOAA Hollings Undergraduate Scholarship
Committee	■ AGU Dewan Scholarship (2026-present) CIRES Members' Council (2025-present) CIRES Mentoring Program (2024-present, Chair) NOAA OAR subject matter expert in <i>Satellite Data</i> (2022-present) University of Miami RSMAS Student Seminar Committee (2016)
Convener/Lead	■ co-Chair , <i>Modeling and Remote Sensing of Earth's Radiation Budget, Climate and Relevant Processes</i> at the AMS's 17 th Conference on Atmospheric Radiation (2026), Madison, WI, USA Chair , <i>Advances in Assessing Aerosol Impacts on Clouds Through AI and Advanced Statistical Techniques</i> , AMS 2026, Houston, TX, USA Discussion Lead , 2025 Radiation and Climate GRS, Lewiston, ME, USA Session co-Lead , University of Washington Inaugural MCB Workshop, 2025, Leavenworth, WA, USA Convener , <i>Advances in Cloud and Precipitation Processes: Integrating Observations, Modeling, and Theory</i> at the 2024 AGU Fall Meeting, Washington D.C., USA co-Chair , <i>Aerosol-Cloud Indirect Effects</i> at the AMS's 16 th Conference on Cloud Physics (2022), Madison, WI, USA
Workshops	■ GEWEX Earth Energy Imbalance Workshop (rapporteur), Pasadena, CA, Jun. 2026 Micro2Macro Workshop by US CLIVAR (rapporteur), Laramie, WY, Oct. 2024 DOE-NOAA Marine Cloud Brightening Workshop (rapporteur), Online, Apr. 2022 EarthCare Workshop, Online, Feb. 2022 Aerosol and Clouds-Convection-Precipitation Workshop, Online, Oct. 2020 NCAR Radiation Workshop, Boulder, CO, Mar. 2016 First DOE ARM Summer School, Norman, OK, Jul. 2015
Peer Reviewer	■ Atmospheric Chemistry and Physics (EGU); Nature Communications, Communications Earth & Environment, npj Climate and Atmospheric Science (Nature), Science Advances (AAAS), Journal of Geophysical Research: Atmosphere, Journal of Geophysical Research: Machine Learning and Computation, Geophysical Research Letters (AGU); Journal of Climate, Bulletin of the American Meteorology Society, Journal of Applied Meteorology and Climatology (AMS); Climate Dynamics (Springer)
Trainings	■ Dialogic Skills Workshop Certificate, University of Colorado Boulder, 2025 Volunteer for CIRES Education & Outreach since 2022 FSU Intramural Basketball Official (2014)
Field campaign	■ Swakopmund, Namibia, NASA ORACLES-2016 , Sept. 2016 São Tomé, São Tomé and Príncipe, NASA ORACLES-2017 , Aug. 2017

Media & Press

- *Understanding Earth's hidden east-west symmetry could improve climate models*, [Link], **Phys.org**, 2026
- *Earth has a mysterious triple symmetry that may influence its climate*, [Link], **New Scientist**, 2026
- *Clouds shape a newly identified east-west symmetry in Earth's reflected sunlight*, [Link], **CIRES News**, 2026
- *Ensuring continuity for atmospheric research*, [Link], **Reflective blog post**, 2025
- *Flying high in the sky*, [Link], **CIRES Spotlights**, 2025
- *The unintended consequences of reducing sulfur emissions from ships*, [Link], **NOAA CPO News**, 2025

- 📖 *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*, [Link], **NOAA Research**, 2025
- 📖 *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*, [Link], **CIRES News**, 2025
- 📖 *Cleaner Ships, Hotter Earth: The Unexpected Climate Twist*, [Link], **SciTechDaily**, 2025
- 📖 *Shipping emissions reduction sheds light on marine cloud geoengineering*, [Link], **Mongabay**, 2024
- 📖 *Smoke Studies: Crucial Cloud-Deck Science*, [Link], **DOE ARM NEWS**, 2021

References

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